

VLSI INTERNSHIP – TASK 2

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Domain: Very Large Scale Integration (VLSI) Intern

Goal:

Synthesize RTL to a gate-level netlist using an academic flow or open-source tool and perform gate-level simulation.

Requirements:

- Use Synopsys DC / Yosys (open-source) for synthesis.
- Generate netlist and perform gate-level simulation with timing if available.
- Compare RTL vs gate-level behavior.

Commands

```
vboxuser@ubuntu:~/alu8bit$ nano synth.js
vboxuser@ubuntu:~/alu8bit$ yosys synth.js
```

```
/-----\
| yosys -- Yosys Open SYnthesis Suite                               |
| Copyright (C) 2012 - 2025  Claire Xenia Wolf <claire@yosyshq.com> |
| Distributed under an ISC-like license, type "license" to see terms |
\-----/
Yosys 0.52 (git sha1 fee39a3284c90249e1d9684cf6944ffbbcbb8f90)

-- Executing script file `synth.js' --

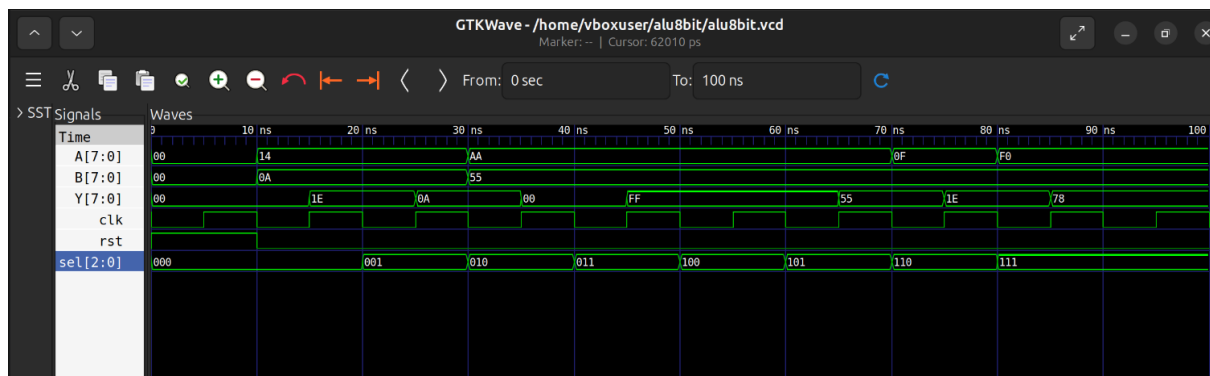
1. Executing Verilog-2005 frontend: alu8bit.v
Parsing Verilog input from `alu8bit.v' to AST representation.
Generating RTLIL representation for module `alu8bit'.
```

```
vboxuser@ubuntu:~/alu8bit$ iverilog -o alu8bit_out2 alu8bit_netlist.v alu8bit_tb.v
```

Synth.js

```
read_verilog alu8bit.v
synth -top alu8bit
write_verilog alu8bit_netlist.v
```

GTKWave after Gate level sythesis



Gate-Level Netlist code:

```
/* Generated by Yosys 0.52 (git sha1 fee39a3284c90249e1d9684cf6944ffbbcbb8f90) */
```

```
(* top = 1 *)
```

```
(* src = "alu8bit.v:1.1-24.10" *)
```

```
module alu8bit(clk, rst, A, B, sel, Y);
```

```
    wire _000_;
```

```
    wire _001_;
```

```
    wire _002_;
```

```
    wire _003_;
```

```
    wire _004_;
```

```
    wire _005_;
```

```
    wire _006_;
```

```
    wire _007_;
```

```
    wire _008_;
```

```
    wire _009_;
```

```
    wire _010_;
```

```
    wire _011_;
```

```
    wire _012_;
```

```
    wire _013_;
```

```
    wire _014_;
```

```
    wire _015_;
```

wire _016_;
wire _017_;
wire _018_;
wire _019_;
wire _020_;
wire _021_;
wire _022_;
wire _023_;
wire _024_;
wire _025_;
wire _026_;
wire _027_;
wire _028_;
wire _029_;
wire _030_;
wire _031_;
wire _032_;
wire _033_;
wire _034_;
wire _035_;
wire _036_;
wire _037_;
wire _038_;
wire _039_;
wire _040_;
wire _041_;
wire _042_;
wire _043_;
wire _044_;

wire _045_;

wire _046_;

wire _047_;

wire _048_;

wire _049_;

wire _050_;

wire _051_;

wire _052_;

wire _053_;

wire _054_;

wire _055_;

wire _056_;

wire _057_;

wire _058_;

wire _059_;

wire _060_;

wire _061_;

wire _062_;

wire _063_;

wire _064_;

wire _065_;

wire _066_;

wire _067_;

wire _068_;

wire _069_;

wire _070_;

wire _071_;

wire _072_;

wire _073_;

wire _074_;
wire _075_;
wire _076_;
wire _077_;
wire _078_;
wire _079_;
wire _080_;
wire _081_;
wire _082_;
wire _083_;
wire _084_;
wire _085_;
wire _086_;
wire _087_;
wire _088_;
wire _089_;
wire _090_;
wire _091_;
wire _092_;
wire _093_;
wire _094_;
wire _095_;
wire _096_;
wire _097_;
wire _098_;
wire _099_;
wire _100_;
wire _101_;
wire _102_;

wire _103_;
wire _104_;
wire _105_;
wire _106_;
wire _107_;
wire _108_;
wire _109_;
wire _110_;
wire _111_;
wire _112_;
wire _113_;
wire _114_;
wire _115_;
wire _116_;
wire _117_;
wire _118_;
wire _119_;
wire _120_;
wire _121_;
wire _122_;
wire _123_;
wire _124_;
wire _125_;
wire _126_;
wire _127_;
wire _128_;
wire _129_;
wire _130_;
wire _131_;

wire _132_;
wire _133_;
wire _134_;
wire _135_;
wire _136_;
wire _137_;
wire _138_;
wire _139_;
wire _140_;
wire _141_;
wire _142_;
wire _143_;
wire _144_;
wire _145_;
wire _146_;
wire _147_;
wire _148_;
wire _149_;
wire _150_;
wire _151_;
wire _152_;
wire _153_;
wire _154_;
wire _155_;
wire _156_;
wire _157_;
wire _158_;
wire _159_;
wire _160_;

wire _161_;
wire _162_;
wire _163_;
wire _164_;
wire _165_;
wire _166_;
wire _167_;
wire _168_;
wire _169_;
wire _170_;
wire _171_;
wire _172_;
wire _173_;
wire _174_;
wire _175_;
wire _176_;
wire _177_;
wire _178_;
wire _179_;
wire _180_;
wire _181_;
wire _182_;
wire _183_;
wire _184_;
wire _185_;
wire _186_;
wire _187_;
wire _188_;
wire _189_;


```
wire _190_;
wire _191_;
wire _192_;
wire _193_;
wire _194_;
wire _195_;
wire _196_;
wire _197_;
wire _198_;
wire _199_;

(* force_downto = 32'd1 *)

(* src = "alu8bit.v:0.0-0.0|alu8bit.v:13.5-22.12|/usr/bin/../share/yosys/techmap.v:576.21-576.22" *)

wire [7:0] _200_;

(* src = "alu8bit.v:4.17-4.18" *)
input [7:0] A;
wire [7:0] A;

(* src = "alu8bit.v:4.20-4.21" *)
input [7:0] B;
wire [7:0] B;

(* src = "alu8bit.v:6.22-6.23" *)
output [7:0] Y;
reg [7:0] Y;

(* src = "alu8bit.v:2.11-2.14" *)
input clk;
wire clk;

(* src = "alu8bit.v:3.11-3.14" *)
input rst;
wire rst;

(* src = "alu8bit.v:5.17-5.20" *)
```

```
input [2:0] sel;
wire [2:0] sel;

assign _130_ = B[0] | A[0];
assign _131_ = B[0] & A[0];
assign _132_ = _130_ & ~(_131_);
assign _133_ = ~sel[2];
assign _134_ = ~(sel[1] & sel[0]);
assign _135_ = _134_ | _133_;
assign _136_ = sel[0] | ~(sel[1]);
assign _137_ = _136_ | _133_;
assign _138_ = _137_ & _135_;
assign _139_ = sel[1] | ~(sel[0]);
assign _140_ = _139_ | _133_;
assign _141_ = sel[1] | sel[0];
assign _142_ = sel[2] & ~(_141_);
assign _143_ = _142_ | ~(_140_);
assign _144_ = _138_ & ~(_143_);
assign _145_ = _133_ & ~(_134_);
assign _146_ = _133_ & ~(_136_);
assign _147_ = _146_ | _145_;
assign _148_ = _144_ & ~(_147_);
assign _149_ = A[1] & ~(_135_);
assign _150_ = ~(_140_ | A[0]);
assign _151_ = ~(B[0] ^ A[0]);
assign _152_ = _142_ & ~(_151_);
assign _153_ = _152_ | _150_;
assign _154_ = _153_ | _149_;
assign _155_ = _145_ & _130_;
assign _156_ = _146_ & _131_;
```

```
assign _157_ = _156_ | _155_;
assign _158_ = _157_ | _154_;
assign _200_[0] = _148_ ? _132_ : _158_;
assign _159_ = ~A[1];
assign _160_ = _133_ & ~(_139_);
assign _161_ = ~(B[0] ^ B[1]);
assign _162_ = ~_161_;
assign _163_ = _160_ ? B[1] : _162_;
assign _164_ = _163_ ^ _159_;
assign _165_ = B[0] & ~(A[0]);
assign _166_ = ~(_165_ ^ _164_);
assign _167_ = A[2] & ~(_135_);
assign _168_ = A[0] & ~(_137_);
assign _169_ = _168_ | _167_;
assign _170_ = _159_ & ~(_140_);
assign _171_ = ~(A[1] ^ B[1]);
assign _172_ = _142_ & ~(_171_);
assign _173_ = _172_ | _170_;
assign _174_ = _173_ | _169_;
assign _175_ = ~(A[1] | B[1]);
assign _176_ = _145_ & ~(_175_);
assign _177_ = ~(A[1] & B[1]);
assign _178_ = _146_ & ~(_177_);
assign _179_ = _178_ | _176_;
assign _180_ = _179_ | _174_;
assign _200_[1] = _148_ ? _166_ : _180_;
assign _181_ = ~A[2];
assign _182_ = ~B[2];
assign _183_ = ~(B[0] | B[1]);
```

```
assign _184_ = _183_ ^ B[2];
assign _185_ = _160_ ? _182_ : _184_;
assign _186_ = _185_ ^ _181_;
assign _187_ = A[1] & ~(_163_);
assign _188_ = _164_ & ~(_165_);
assign _189_ = _188_ | _187_;
assign _190_ = ~(_189_ ^ _186_);
assign _191_ = A[3] & ~(_135_);
assign _192_ = A[1] & ~(_137_);
assign _193_ = _192_ | _191_;
assign _194_ = _181_ & ~(_140_);
assign _195_ = ~(A[2] ^ B[2]);
assign _196_ = _142_ & ~(_195_);
assign _197_ = _196_ | _194_;
assign _198_ = _197_ | _193_;
assign _199_ = ~(A[2] | B[2]);
assign _000_ = _145_ & ~(_199_);
assign _001_ = ~(A[2] & B[2]);
assign _002_ = _146_ & ~(_001_);
assign _003_ = _002_ | _000_;
assign _004_ = _003_ | _198_;
assign _200_[2] = _148_ ? _190_ : _004_;
assign _005_ = ~A[3];
assign _006_ = ~B[3];
assign _007_ = _183_ & ~(B[2]);
assign _008_ = _007_ ^ B[3];
assign _009_ = _160_ ? _006_ : _008_;
assign _010_ = _009_ ^ _005_;
assign _011_ = ~(_185_ & A[2]);
```

```
assign _012_ = _189_ & ~(_186_);
assign _013_ = _011_ & ~(_012_);
assign _014_ = _013_ ^ _010_;
assign _015_ = A[4] & ~(_135_);
assign _016_ = A[2] & ~(_137_);
assign _017_ = _016_ | _015_;
assign _018_ = _005_ & ~(_140_);
assign _019_ = ~(A[3] ^ B[3]);
assign _020_ = _142_ & ~(_019_);
assign _021_ = _020_ | _018_;
assign _022_ = _021_ | _017_;
assign _023_ = ~(A[3] | B[3]);
assign _024_ = _145_ & ~(_023_);
assign _025_ = ~(A[3] & B[3]);
assign _026_ = _146_ & ~(_025_);
assign _027_ = _026_ | _024_;
assign _028_ = _027_ | _022_;
assign _200_[3] = _148_ ? _014_ : _028_;
assign _029_ = ~A[4];
assign _030_ = ~B[4];
assign _031_ = B[3] | B[2];
assign _032_ = _183_ & ~(_031_);
assign _033_ = _032_ ^ B[4];
assign _034_ = _160_ ? _030_ : _033_;
assign _035_ = _034_ ^ _029_;
assign _036_ = ~(_009_ & A[3]);
assign _037_ = ~(_011_ | _010_);
assign _038_ = _036_ & ~(_037_);
assign _039_ = _010_ | _186_;
```

```
assign _040_ = _189_ & ~(_039_);
assign _041_ = _038_ & ~(_040_);
assign _042_ = _041_ ^ _035_;
assign _043_ = A[5] & ~(_135_);
assign _044_ = A[3] & ~(_137_);
assign _045_ = _044_ | _043_;
assign _046_ = _029_ & ~(_140_);
assign _047_ = ~(A[4] ^ B[4]);
assign _048_ = _142_ & ~(_047_);
assign _049_ = _048_ | _046_;
assign _050_ = _049_ | _045_;
assign _051_ = ~(A[4] | B[4]);
assign _052_ = _145_ & ~(_051_);
assign _053_ = ~(A[4] & B[4]);
assign _054_ = _146_ & ~(_053_);
assign _055_ = _054_ | _052_;
assign _056_ = _055_ | _050_;
assign _200_[4] = _148_ ? _042_ : _056_;
assign _057_ = ~A[5];
assign _058_ = ~B[5];
assign _059_ = _032_ & ~(B[4]);
assign _060_ = _059_ ^ B[5];
assign _061_ = _160_ ? _058_ : _060_;
assign _062_ = _061_ ^ _057_;
assign _063_ = ~(_034_ & A[4]);
assign _064_ = ~(_041_ | _035_);
assign _065_ = _063_ & ~(_064_);
assign _066_ = _065_ ^ _062_;
assign _067_ = A[6] & ~(_135_);
```

```

assign _068_ = A[4] & ~(_137_);
assign _069_ = _068_ | _067_;
assign _070_ = _057_ & ~(_140_);
assign _071_ = ~(A[5] ^ B[5]);
assign _072_ = _142_ & ~(_071_);
assign _073_ = _072_ | _070_;
assign _074_ = _073_ | _069_;
assign _075_ = ~(A[5] | B[5]);
assign _076_ = _145_ & ~(_075_);
assign _077_ = ~(A[5] & B[5]);
assign _078_ = _146_ & ~(_077_);
assign _079_ = _078_ | _076_;
assign _080_ = _079_ | _074_;
assign _200_[5] = _148_ ? _066_ : _080_;
assign _081_ = ~B[6];
assign _082_ = B[5] | B[4];
assign _083_ = _032_ & ~(_082_);
assign _084_ = _083_ ^ B[6];
assign _085_ = _160_ ? _081_ : _084_;
assign _086_ = _085_ ^ A[6];
assign _087_ = ~(_061_ & A[5]);
assign _088_ = ~(_063_ | _062_);
assign _089_ = _087_ & ~(_088_);
assign _090_ = _062_ | _035_;
assign _091_ = ~(_090_ | _041_);
assign _092_ = _089_ & ~(_091_);
assign _093_ = ~(_092_ ^ _086_);
assign _094_ = A[7] & ~(_135_);
assign _095_ = A[5] & ~(_137_);

```

```
assign _096_ = _095_ | _094_;
assign _097_ = ~(_140_ | A[6]);
assign _098_ = ~(A[6] ^ B[6]);
assign _099_ = _142_ & ~(_098_);
assign _100_ = _099_ | _097_;
assign _101_ = _100_ | _096_;
assign _102_ = ~(A[6] | B[6]);
assign _103_ = _145_ & ~(_102_);
assign _104_ = ~(A[6] & B[6]);
assign _105_ = _146_ & ~(_104_);
assign _106_ = _105_ | _103_;
assign _107_ = _106_ | _101_;
assign _200_[6] = _148_ ? _093_ : _107_;
assign _108_ = ~A[7];
assign _109_ = ~B[7];
assign _110_ = _083_ & ~(B[6]);
assign _111_ = _110_ ^ B[7];
assign _112_ = _160_ ? _109_ : _111_;
assign _113_ = _112_ ^ _108_;
assign _114_ = ~(_085_ & A[6]);
assign _115_ = _086_ & ~(_092_);
assign _116_ = _114_ & ~(_115_);
assign _117_ = _116_ ^ _113_;
assign _118_ = A[6] & ~(_137_);
assign _119_ = _108_ & ~(_140_);
assign _120_ = ~(A[7] ^ B[7]);
assign _121_ = _142_ & ~(_120_);
assign _122_ = _121_ | _119_;
assign _123_ = _122_ | _118_;
```



```

assign _124_ = ~(A[7] | B[7]);
assign _125_ = _145_ & ~(_124_);
assign _126_ = ~(A[7] & B[7]);
assign _127_ = _146_ & ~(_126_);
assign _128_ = _127_ | _125_;
assign _129_ = _128_ | _123_;
assign _200_[7] = _148_ ? _117_ : _129_;
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[0] <= 1'h0;
    else Y[0] <= _200_[0];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[1] <= 1'h0;
    else Y[1] <= _200_[1];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[2] <= 1'h0;
    else Y[2] <= _200_[2];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[3] <= 1'h0;
    else Y[3] <= _200_[3];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[4] <= 1'h0;
    else Y[4] <= _200_[4];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)

```

```
    if (rst) Y[5] <= 1'h0;
    else Y[5] <= _200_[5];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[6] <= 1'h0;
    else Y[6] <= _200_[6];
(* src = "alu8bit.v:9.1-22.12" *)
always @(posedge clk, posedge rst)
    if (rst) Y[7] <= 1'h0;
    else Y[7] <= _200_[7];
endmodule
```